

Mostafa Ashraf Mohamed

Personal Info

Phone: +20 111 223 2652
Military Service: Completed

Email: mostaphaashraf1996@gmail.com
Address: Cairo, Egypt

[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

Education

Master of Science (M.Sc.) in Electrical Engineering (Computer & Systems Engineering) Ain Shams University (ASU), Faculty of Engineering | **2024 – 2026**

Bachelor of Engineering (B.Sc.) in Electrical Engineering (Computer & Systems Engineering) Ain Shams University (ASU) | **Graduated: 2021**

Graduation Project: Brain-Controlled Smart Wheelchair - [BrainWheel](#) — **Grade:** Excellent

- EEG-based control using machine learning for movement prediction.
- Autonomous navigation mode with safety constraints.
- **Ranked Top 5** in Dell's Envision the Future competition, **sponsored** by Valeo and iHub.

Work Experience

National Telecommunication Institute (NTI)	Instructor , Cairo, Egypt - Teach core software development fundamentals for multiple technical tracks. - Delivered specialized training in Applied Data Science, DevOps, and Cloud. - Prepared candidates for AWS certifications, including: - AWS Cloud Practitioner, AWS Developer, AWS Solutions Architect - Designed course materials, hands-on labs, and real-world use cases.	Jun 2025 - Present
Orange Innovation	Software Engineer , (C++ - Qt - Cloud - CI/CD - 5G ORAN), - Designed and implemented a custom communication protocol over LoRa mesh networks, following OSI layered architecture. - Developed Qt-based desktop applications for protocol control, visualization, and monitoring. - Contributed to ns-O-RAN / FlexRIC open-source projects: - Implemented E2AP v1.01 and KPM-RC v3 support. - Developed KPI-based handover decision logic. - Implemented an Energy Saving xApp for 5G RAN optimization. - Integrated RIC-TaaP with FlexRIC by OpenAirInterface (OAI) for LTE/5G simulation and testing. - Presented Energy Saving xApp at OAI 10th Anniversary Workshop (France) - Currently developing a Charging Function (CHF) for traffic calculation and quota management in OAI Core, Docker, IaC, CI/CD. - Conducted code reviews, refactored legacy components, and mentored interns.	Apr 2022 - Present
Faculty of Engineering, (ASU)	Teaching Assistant , Ain Shams University, Cairo, Egypt - Taught Distributed Computing, Data Structures & Algorithms, High-Performance Computing, and Advanced Programming. - Developed course materials, labs, activities, and supervised student projects.	Feb 2022 - Oct 2025
Government Service	Full Stack Software Developer , Cairo, Egypt - Built full-stack applications using ReactJS, NodeJS, and SQL databases. - Integrated REST APIs and WebSockets to connect embedded devices with UI. - Configured a local network for 50+ PCs using routers and switches.	Nov 2022 - Nov 2023
Google Summer of Code 2023	Mentor at Jenkins - Guided contributors on effective code reviews, documentation, and feedback. - Assisted in integrating their projects into the codebase.	Feb 2023 - Aug 2023
Google Summer of Code 2022	Open Source Contributor at CGAL , (C++11/17, Qt, OpenGL) - Refactored the CGAL open-source computational geometry library following the Model–View–Presenter (MVP) design pattern. - Enhanced the library to support runtime configurable features and UI integration. - Applied template metaprogramming, added new features, fixed bugs, and enabled OpenGL support.	May 2022 - Oct 2022

Google Summer of Code 2021	Open Source Contributor at GraphicsFuzz, (C++, Spir-v) May 2021 - Aug 2021 Provided support and contributions to SPIRV-Fuzz, a tool used to explore vulnerabilities in GPU drivers on Android and PC platforms. Applied metamorphic testing principles to generate diverse shader inputs and uncover subtle driver and compiler issues. Contributed to Shadertrap by developing and integrating GLSL shaders into the Google codebase. Designed GPU shader programs executed directly on the GPU. - Identified and reported bugs in SwiftShader and spirv-opt.
COM-IoT Technologies	Software Engineering Intern, (C++, Qt) Feb 2021 - Apr 2021 Developed a Kalman filter algorithm used in autonomous vehicles.
Resala RTC	Instructor, Cairo, Egypt Jun 2018 - Aug 2020 Taught C++, object-oriented programming, data structures, and algorithms.
Sphinx and CSSI Factory	Interior Design, Industrial Zone, October, Egypt Jun 2016 - Jan 2017 AutoCAD drafter, document controller

Skills

- **Languages:** C, C++11/14/17, Java, Python, JavaScript.
- Object-Oriented Design, Design patterns, Multithreading, Concurrency, Memory Management, Data Structures, Algorithms, SOLID principles, Networks, Operating Systems, Distributed Systems, Databases(SQL, NoSQL), Microservices, Linux, and Agile Scrum.
- **Frameworks & Tools:** Qt, CMake, GDB, Valgrind, ASAN, TSAN, Git, gRPC, Docker, Kubernetes, CI/CD, ReactJS, NodeJS, and MongoDB, Grafana, Terraform.
- **Libraries & concepts:** OpenMP, MPI, Numpy, Pandas, SoA, SOAP, and RESTful APIs
- **Cloud:** AWS services (EC2, Lambda, S3, SageMaker)
- **GPU Development:** CUDA programming. Basics of electronic devices and circuits, VLSI, and VHDL.
- Architectural and AutoCAD drafter, presentation, open source contributor, and problem-solving skills.

Projects

- [RIC Testing as a Platform \(ns-O-RAN\)](#) - Open-source testing platform for xApps and RIC integration using OAI and ns-3. **C++, Linux, Docker, Grafana, InfluxDB**
- [OS Simulator](#) - Implemented memory management, scheduling algorithms. **C++11, Qt, Linux**
- [Linux System Monitor \(htop-like\)](#) - Process and resource monitoring tool. **C++, Linux**
- [Search Engine](#) - Indexed 100K+ files using **Trie** indexing, optimized for fast search. **C++, Qt, PostgreSQL**
- [Machine Learning Projects](#) - CNN Dog Breed Classifier, Plagiarism Detector, Sentiment Analysis Web App, and [TorchPIP](#) (Basic Deep Learning framework). **Python, Flask, AWS**

Courses

Cloud DevOps Nanodegree, Udacity, August 2022	AWS re/Start BootCamp, Sprints, July 2022
Complete Modern C++ (C++11/14/17), Dec 2021	AWS Machine Learning Certified, MCIT, August 2021
React Developer Nanodegree, Udacity, May 2021	Machine Learning Nanodegree, Udacity, Dec 2020
C++ Nanodegree Program, Udacity, July 2020	Software Design Specialization, Coursera, June 2020
Python Specialization, Coursera, NTL, May 2020	Embedded Systems diploma, 150 hours, May 2018

Certifications

- **AWS Certified Cloud Practitioner**, [CLF-C02](#), August 2024
- Coding4Integrity, UN(United Nations) Competition, Oct 2021
- 2nd place at local ACM, ACPC Problem Solving Competition, Sep 2019

Extracurricular Activities

DFA Learning Deutsch	Sep 2021 - Feb 2022	APEC IT Head	Sep 2020 - May 2021
Udacity Student Ambassador	Nov 2020 - Mar 2021	TIEC Entrepreneurship Trainee	Aug 2020 - Sep 2020

Language

Arabic, Native	English, C1	German, A1
----------------	-------------	------------